IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Executive Summary

- I obtained data on historic launches using the SpaceX REST API, complemented by parsing Space X launch data from Wikipedia. These data were cleaned and formatted before incorporation into a dataframe. Explanatory figures of historic trends were generated using Pandas functions. Geographic trends were identified using Folium maps, and a Plotly dashboard was created to compare site-relevant data using interactive pie charts and scatter plots. Finally predictive analysis through testing several machine learning algorithms to determine the best model predictor of booster reuse by regression.
- I find launches take place from coastal launch sites in the continental United States. Success in first stage landing (enabling booster reuse) has increased over time and is highest for payloads in the 3,000 to 4,000 kg range. Launches to higher orbits or higher payload mass result in a lower probability of booster reuse. A decision tree classifier predictive model is found to be best for first stage reuse with an accuracy of 88%.

Introduction

- Space Y, founded by billionaire industrialist Allon Musk, would like to compete with rocket company SpaceX. The cost of a rocket's first stage is a substantial fraction of the launch cost; Space X's Falcon 9 has a price tag of $62mn per launch compared to $165mn for competitors. Determining under what circumstances a first stage could be reused would identify significant cost savings for Space Y.
- In this project I was tasked with determining the price of each Space X launch and further determine if Space X would reuse the first stage by identifying successful landings through predictive modelling.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

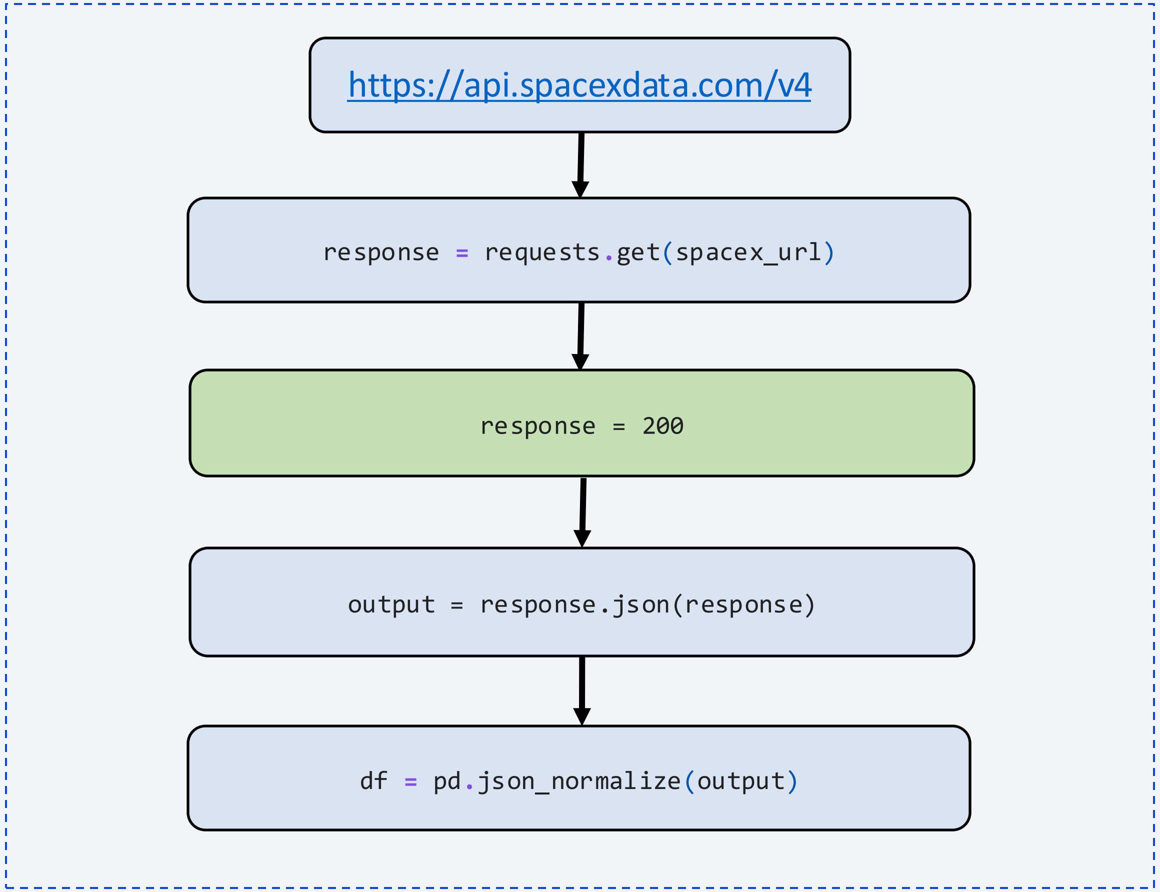
Data Collection

- Rocket launch data were obtained using the Python requests module from a static JSON object.
- The request response was interpreted using the .json() method then converted into a Pandas data frame using the .json_normalize() method.
- Columns 'failures' and 'cores' with dictionary data were split into subcolumns processing the column values element-wise.
- The data frame was then restricted to relevant columns, list items converted to single value entries, and the data frame trimmed by date range.
- At this point the SpaceX API is no longer accessible so the 'rocket', 'payload', 'launch_site', and 'cores' columns could not be queried to get the required information.
- Acquired data would have been combined into a new data frame.
- The new data frame would have been filtered to remove non-Falcon 9 launches from the data set and replace missing payload mass values with the mean of that column.

Data Collection – SpaceX API

- Request rocket launch data from SpaceX API at given URL using `requests`. Check for valid response ('200'). Parse response as a JSON object using `.json()`. Convert JSON into a Pandas dataframe using `.json_normalize()`.

- https://github.com/jontymarshall/IBMDataScienceCapstoneProject/blob/main/Capstone_Data_Collection_API.ipynb



Data Collection - Scraping

- Obtain rocket launch data from Wikipedia URL using `requests`. Check for valid response ('200'). Parse response as html using `BeautifulSoup`. Extract table using `.find_all('table')`. Extract column names from table using `.find_all('th')` to identify table headers and convert list into a dictionary using `dict.fromkeys()`. Iterate over table rows to extract values to populate empty `dicts` with column names, then combine `dicts` to create data frame.

- https://github.com/jontymarshall/IBMDataScienceCapstoneProject/blob/main/Capstone_Webscraping.ipynb

```
https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922
↓
output = requests.get(static_url,headers=headers)
↓
response = 200
↓
soup = BeautifulSoup(output.text,'html.parser')
↓
html_tables = soup.find_all('table')
first_launch_table = html_tables[2]
↓
colheads = first_launch_table.find_all('th')
column_names.append(extract_column_from_header(text)
↓
launch_dict= dict.fromkeys(column_names)
↓
df= pd.DataFrame({ key:pd.Series(value) for key, value in launch_dict.items() })
```

Data Wrangling

- Data were ingested from a csv file into a pandas data frame. Ingestion was assessed by printing the data frame header and printing the column data types. The amount of missing data for each column was then calculated from the fraction of null values. The number of launch sites and the types of orbit and mission outcomes were calcuated using the value_counts() method on the relevant columns. Bad outcomes were identified as a dictionary object and a binary integer outcome 'Class' calculated by element-wise comparison of the 'Outcome' column to the dictionary contents.
- https://github.com/jontymarshall/IBMDataScienceCapstoneProject/blob/main/Capstone_Data_Wrangling.ipynb

EDA with Data Visualization

- Summarize charts and why you used them:
  - Payload mass vs. Flight number -- *Payload mass increases with flight number.*
  - Launch site vs. Flight number -- *Success generally increases over time for all launch sites. VAFB has many fewer launches spread across the full range of flight numbers whereas the CCAFS and KFC are more heavily used, swapping between the two.*
  - Launch site vs. Payload mass -- *VAFB-SLC launch site has no payload masses above 10,000 kg, the others host launches with a broad range of payload masses.*
  - Orbit vs. Flight number -- *Launch success to most orbits increases with flight number except GTO where outcomes are mixed.*
  - Class vs. Orbit – *Identify success rate vs. Orbit types of which ES-L1, GEO, HEO, and SSO have the highest success rates.*
  - Payload mass vs. Orbit -- *High mass payloads are more successful for LEO, ISS, and PO type orbits. GTO orbit has very mixed outcomes.*
  - Success rate vs. Year -- *Launch success rate increases from 2013 to 2020.*
- https://github.com/jontymarshall/IBMDataScienceCapstoneProject/blob/main/Capstone_EDA_with_Visualisation.ipynb

EDA with SQL

- Summary of SQL queries:

```
1. %sql select distinct "Launch_Site" from SPACEXTABLE; -- Identify distinct launch sites in the data set

2. %sql select * from SPACEXTABLE where "Launch_Site" like "CCA%" limit 5; -- Return five launches from sites starting with CCA

3. %sql select SUM("PAYLOAD_MASS__KG_") from SPACEXTABLE where "Customer" = "NASA (CRS)"; -- Total payload mass launched for NASA

4. %sql select AVG("PAYLOAD_MASS__KG_") from SPACEXTABLE where "Booster_Version" = "F9 v1.1"; -- Average payload mass launched by Falcon 9 v1.1 booster variant

5. %sql select MIN("Date") from SPACEXTABLE where "Mission_Outcome" = "Success" and "Landing_Outcome" = "Success (ground pad)"; -- Earliest date of successful
   landing on a ground pad

6. %sql select "Booster_Version" from SPACEXTABLE where "Landing_Outcome" = "Success (drone ship)" and "PAYLOAD_MASS__KG_" between 4000 and 6000; -- List names of
   boosters with payloads between 4000 and 6000Kg and successful drone ship landings

7. %sql select "Mission_Outcome", count(*) from SPACEXTABLE group by "Mission_Outcome"; -- List total successful and failed mission outcomes

8. %sql select distinct("Booster_Version") from SPACEXTABLE where "PAYLOAD_MASS__KG_" = (select MAX("PAYLOAD_MASS__KG_") from SPACEXTABLE); -- List booster
   versions that have carried the maximum payload mass

9. %sql select Booster_Version, Launch_Site, Landing_Outcome, substr(Date, 0, 5) as Year, substr(Date, 6,2) as Month from SPACEXTABLE where Year = '2015'; -- List
   outcomes for launches that took place in 2015

10. %sql select Landing_Outcome, count(Landing_Outcome) as Number from SPACEXTABLE where Date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order
    by Number desc; -- List count of landing outcomes between 2010-06-04 and 2017-03-20 in order of descending frequency
```

- https://github.com/jontymarshall/IBMDataScienceCapstoneProject/blob/main/Capstone_EDA_with_SQL.ipynb

Build an Interactive Map with Folium

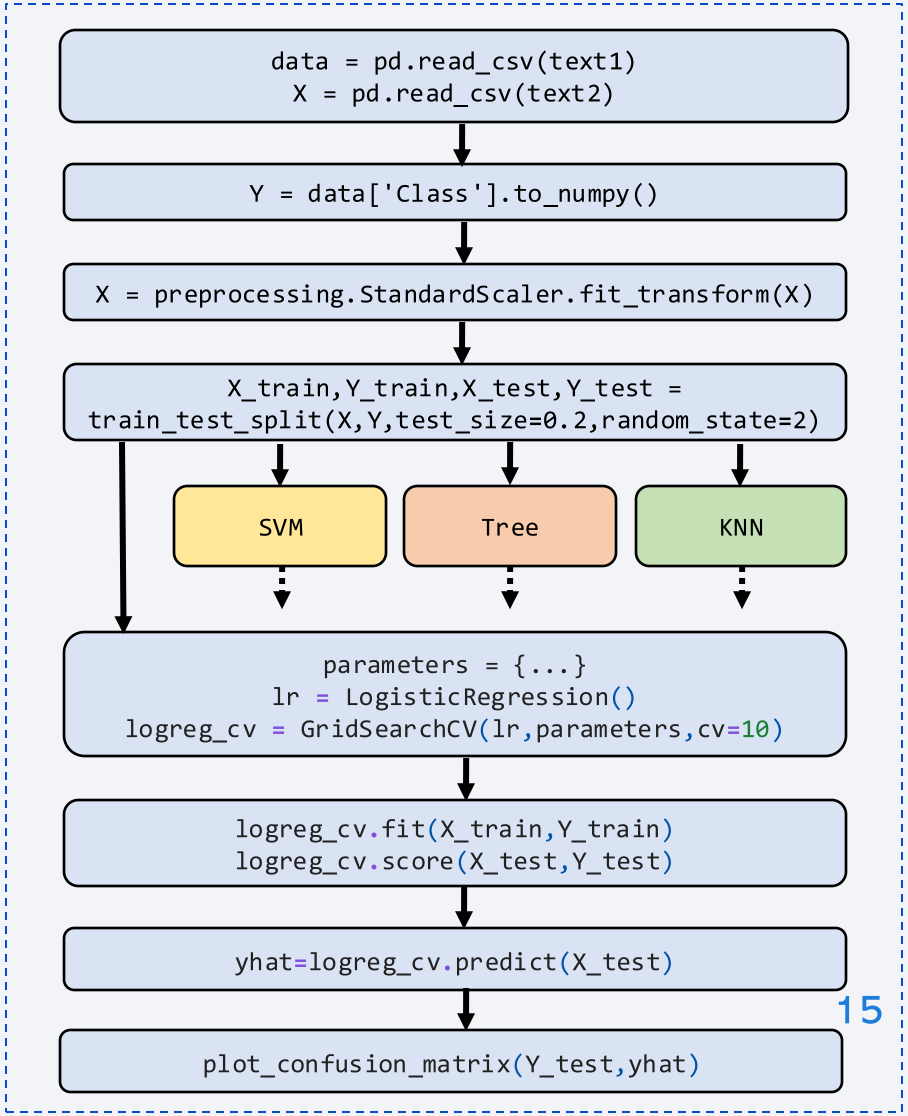
- Map objects created and added to a folium map:
  - Circle and Marker objects added for each launch site position
  - Marker_Cluster objects added for each launch site position containing coloured markers denoted successful and failed launches from that site
  - PolyLine objects added from a launch site to nearby locations of interest (city, coastline, highway, railway)
- The circle and marker objects for each launch site were added to determine commonalities in their locations. The marker_cluster objects contained the results of launches from each site to visualize the density of launches and their success rates. The polyline objects visualize the relationship between each launch site and its environment including geography, supporting infrastructure, and population centres.
- https://github.com/jontymarshall/IBMDataScienceCapstoneProject/blob/main/Capstone_Folium.ipynb

Build a Dashboard with Plotly Dash

- Summary of plots/graphs and interactions added to the dashboard:
    - Dropdown menu for site selection
    - Pie chart plot object to present launch success (+ failure) for the selected launch site
    - Payload mass slider object to select a range of payloads to plot on the scatter graph
    - Scatter plot graph object to show payload mass vs success/failure for the selected launch site and payload mass range
- The dropdown menu for site selection filters the complete data set by location (either "all" or an individual site) to be summarised in the dashboard. The pie chart shows the number of successes per site (for "all") or success and failures (for individual site). The payload mass slider further filters the data set by limiting the range of payload mass launch outcomes to be presented for the selected site. The final scatter plot shows the success/failure outcomes for the selected site illustrating the outcome by location and launch mass combined.
- https://github.com/jontymarshall/IBMDataScienceCapstoneProject/blob/main/Capstone_Plotly_Dash_App.py

Predictive Analysis (Classification)

- Loaded data in pandas dataframes 'data' and 'X' using `pd.from_csv()`. Extracted 'class' column from 'data', converted to numpy array and assigned as the target ('Y') variable. Transformed 'X' feature set using `preprocessing.StandardScaler()`. Split the data into training and testing sets using `train_test_split()`. Used `GridSearchCV()` to create four regression method objects (logistic regression, support vector machine, decision tree, K nearest neighbours). Fitted the regression methods on training data. Calculated accuracy, score, and predictions for each regression method. Finally plotted confusion matrices for all methods.
- https://github.com/jontymarshall/IBMDataScienceCapstoneProject/blob/main/Capstone_Predictive_Analysis.ipynb



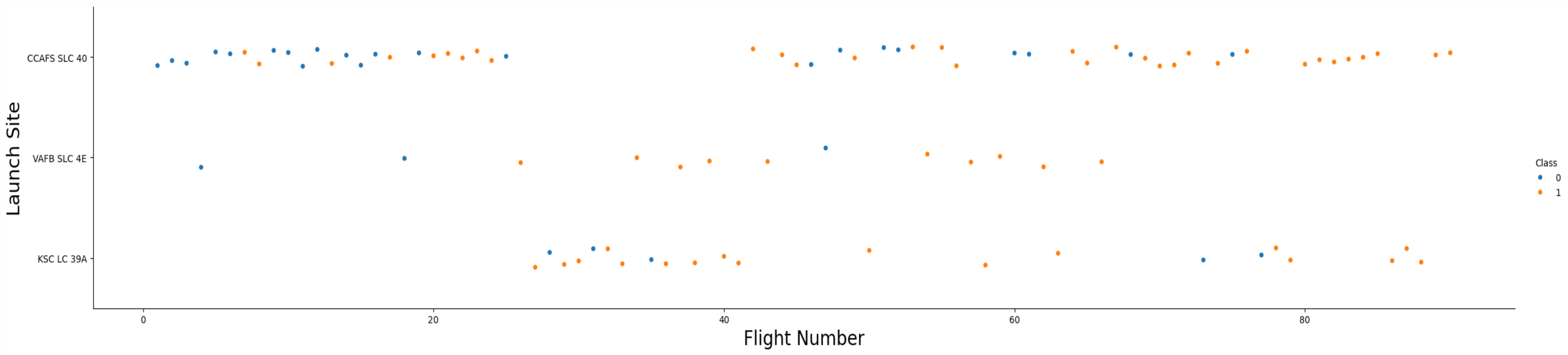
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

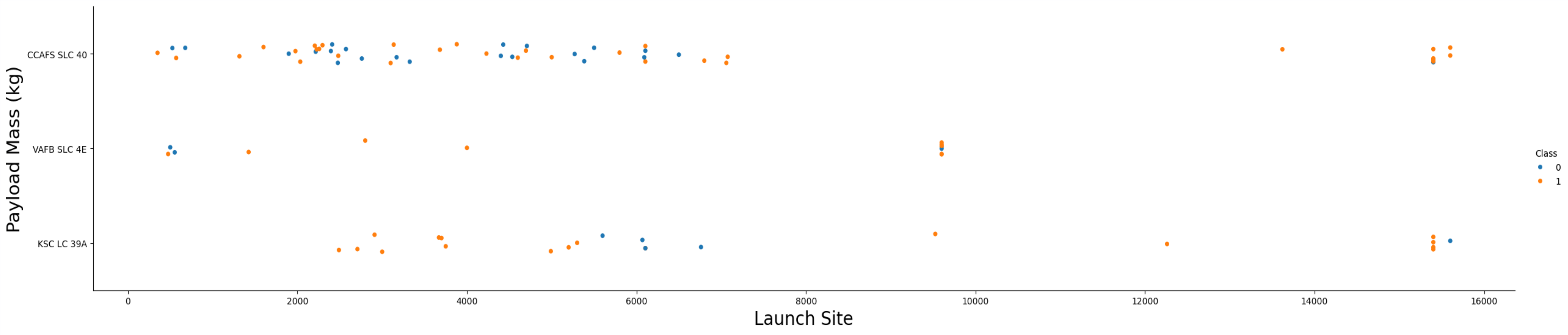
Insights drawn from EDA

Flight Number vs. Launch Site



- Scatter plot of Flight Number vs. Launch Site, data points are coloured by 'class' where 0 (blue) denotes a launch failure and 1 (orange) denotes a launch success.
- Launches predominantly occur from the East coast (CCAFS and KSC). Earliest launches took place from CCAFS, before switching to KSC then back to CCAFS. Intermittent launches take place from VAFB suggesting there is something special about these payloads.

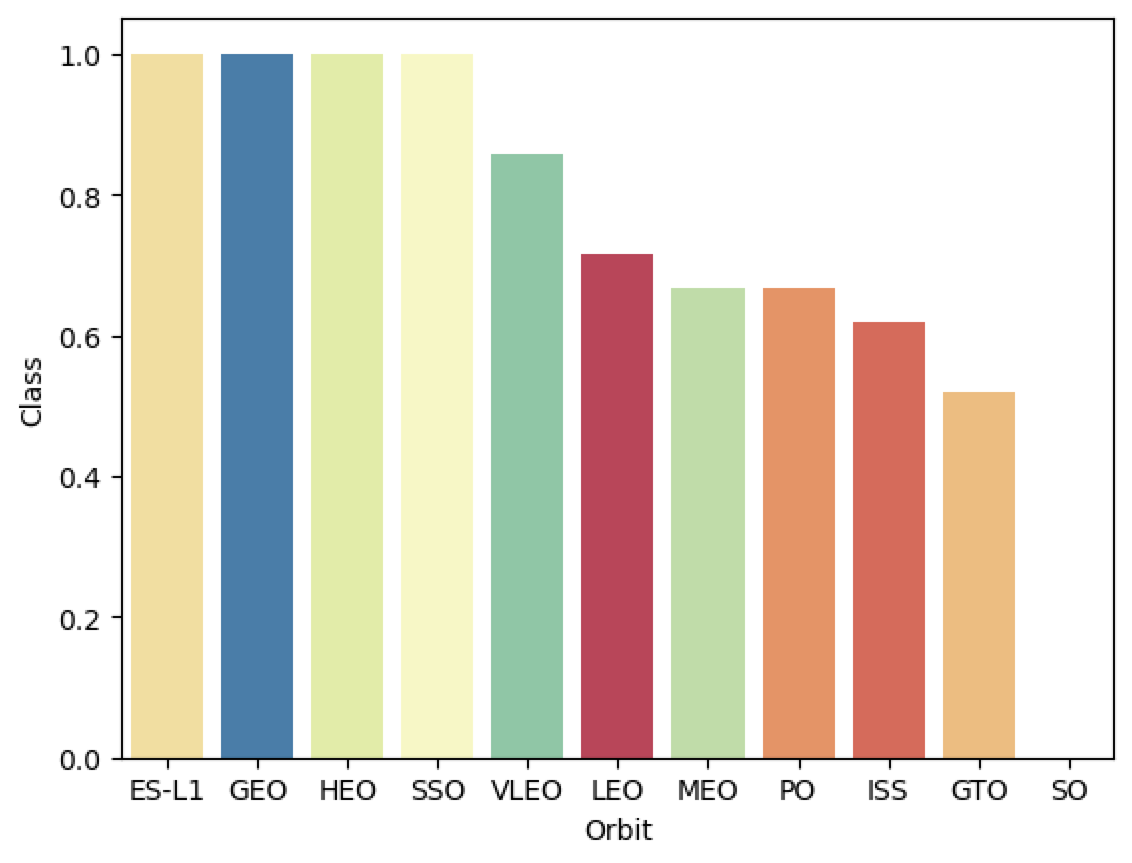
Payload vs. Launch Site



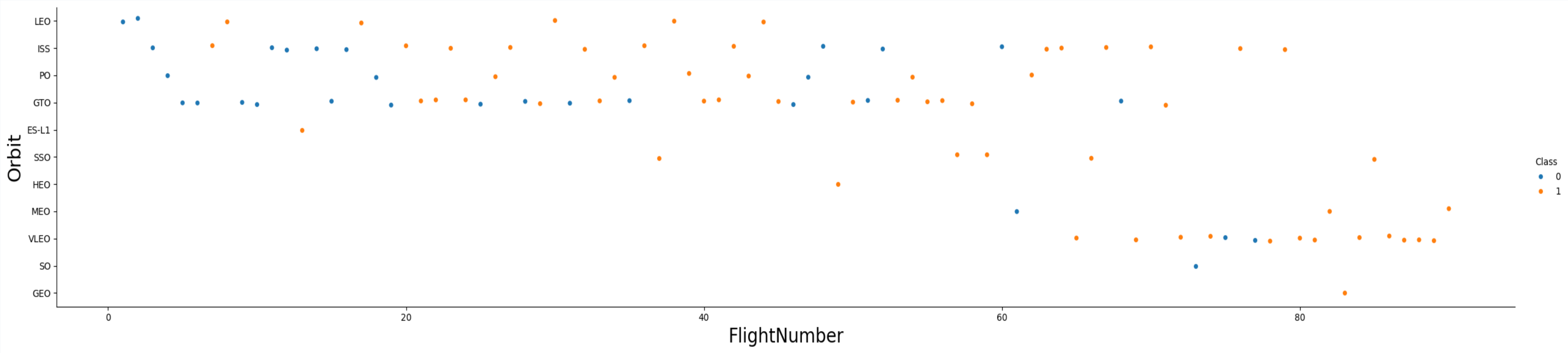
- Scatter plot of Payload vs. Launch Site, data points are coloured by 'class' where 0 (blue) denotes a launch failure and 1 (orange) denotes a launch success.
- CCAFS SLC 40 is the most used launch site. Launch success correlates with payload mass across all sites, with high (> 8000 kg) payload masses being more likely successful compared to lower payload masses.

Success Rate vs. Orbit Type

- Bar chart for the fractional success rate of launches versus orbit type.
- All orbits have a success rate >= 50%. GTO has the lowest success rate for any orbit, pointing to the complexity of in-orbit maneuvers. Orbits with 100% success rates, ES-L1 GEO HEO and SSO, have few launches (< 5 for each).

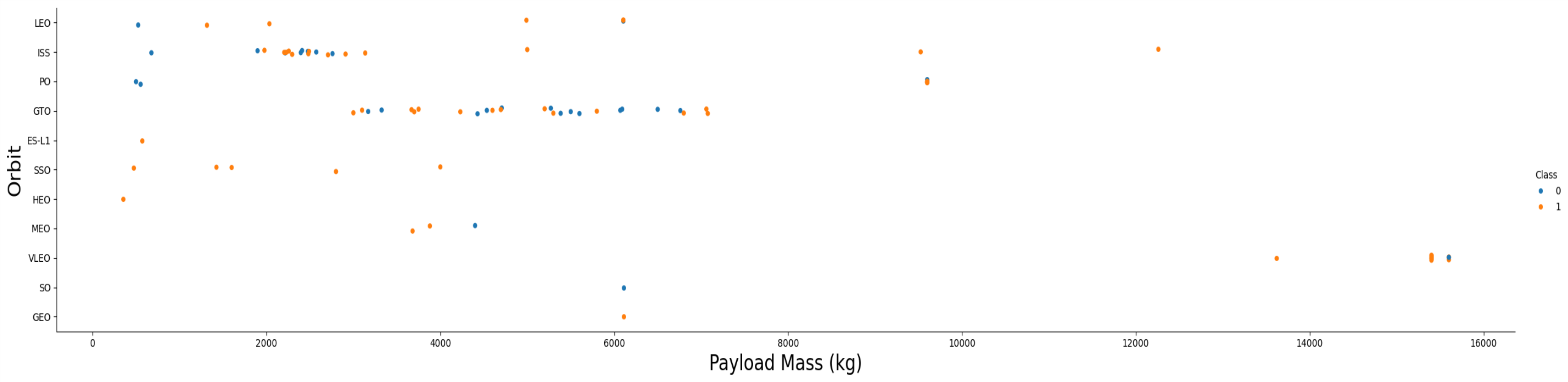


Flight Number vs. Orbit Type



- Scatter plot of Flight number vs. Orbit type for launches, data points are coloured by 'class' where 0 (blue) denotes a launch failure and 1 (orange) denotes a launch success.
- The likelihood of launch success has increased over time, with very low probability before launch #20 (5/20). Higher orbits (M/HEO, SO, GEO) are targeted for launches as time goes by. GTO has the greatest overall statistical chance of launch failure (13/27 launches).

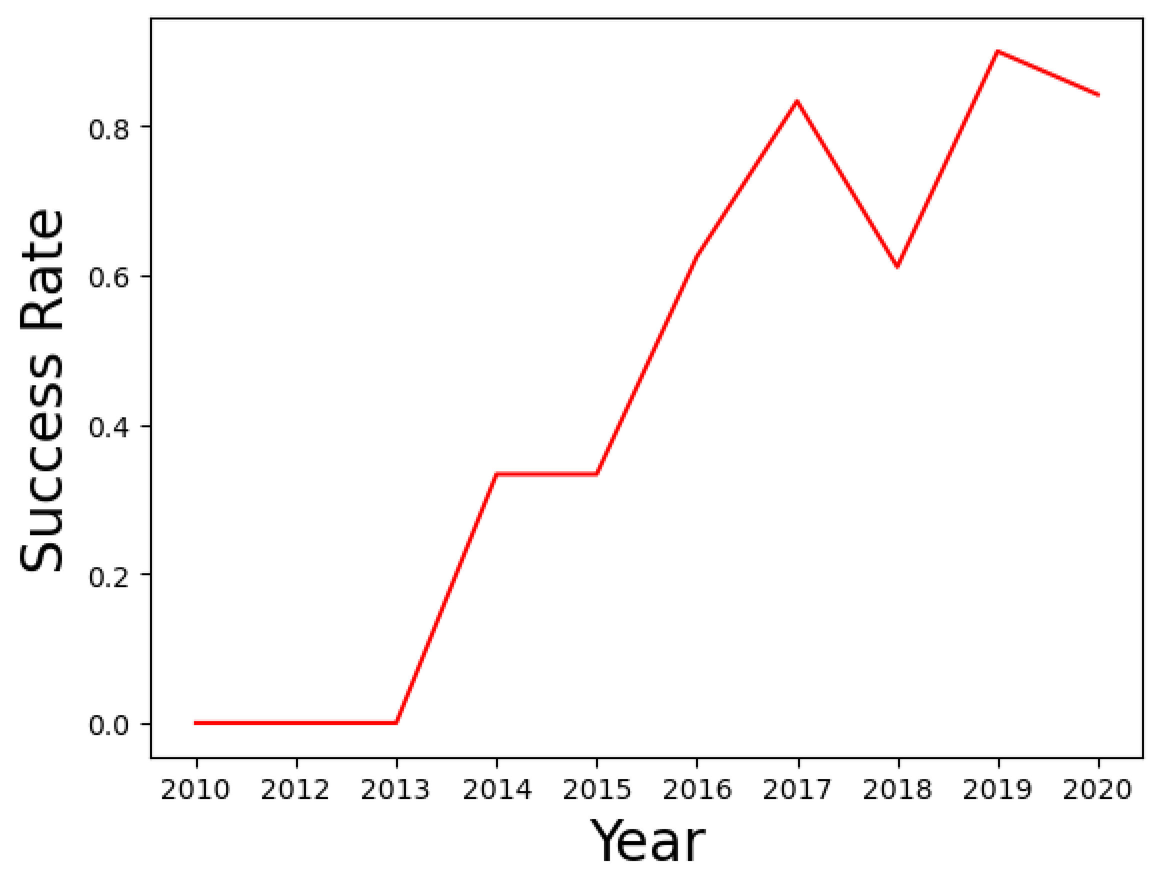
Payload vs. Orbit Type



- Scatter plot of Payload vs. Orbit type, data points are coloured by 'class' where 0 (blue) denotes a launch failure and 1 (orange) denotes a launch success.
- Payload masses > 8000 kg have highest success. Payloads to ISS are clustered around 2000-3000 kg, and payloads to GTO are clustered around 3000-7000 kg. The orbits requiring highest payloads are VLEO, PO, and the ISS, with maxima above 8000 kg.

Launch Success Yearly Trend

- Yearly average (fractional) success rate denoted by the blue solid line, with the standard deviation of the success rate denoted by the blue shaded region.
- The success rate is initially 0% until 2013, and increases ~linearly over time reaching ~80% by 2020.



All Launch Site Names

- Unique launch site names in the data set are: 'CCAFS SLC 40', 'KSC LC 39A', and 'VAFB SLC 4E'

- To obtain this result I used the query:
  `nlaunch = df.value_counts('LaunchSite')`

  which returned the number of each instance of unique names in the dataframe's 'LaunchSite' column
  `LaunchSite`

- `CCAFS SLC 40     55`
- `KSC LC 39A       22`
- `VAFB SLC 4E      13`

Launch Site Names Begin with 'CCA'

- I used the following query to find 5 records where launch sites begin with `CCA`

```
%sql select * from SPACEXTABLE where "Launch_Site" like "CCA%" limit 5;
```

- This query selects a maximum of five records from the table filtering the launch site to contain the text "CCA%"

- Query result:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- I used the following query to calculate the total payload carried by boosters from NASA:

```
%sql select SUM("PAYLOAD_MASS__KG_") from SPACEXTABLE where "Customer"
= "NASA (CRS)";
```

- This query takes the total (sum) of the payload mass column where the customer name was NASA.

- Query result:

SUM("PAYLOAD_MASS__KG_")
45596

Average Payload Mass by F9 v1.1

- I used the following query to identify the average payload mass carried by the F9 v1.1 booster:

```
%sql select AVG("PAYLOAD_MASS__KG_") from SPACEXTABLE where
"Booster_Version" = "F9 v1.1";
```

- This query takes the average of the payload masses column filtered by the booster version to select only 'F9 v1.1'.

- Query result:

AVG("PAYLOAD_MASS__KG_")
2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

```
%sql select MIN("Date") from SPACEXTABLE where "Mission_Outcome" = "Success" and "Landing_Outcome" = "Success (ground pad)"
```

- I used the minimum function on the date column where the mission outcome was success and landing outcome was success (ground pad) to filter the results.

- Query result:

MIN("Date")
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- I used the following query to identify names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql select "Booster_Version" from SPACEXTABLE where
"Landing_Outcome" = "Success (drone ship)" and "PAYLOAD_MASS__KG_"
between 4000 and 6000;
```

- This returns the booster versions that landed on a drone ship with a payload in the 4000 to 6000 kg range.

- Query result:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- I used the following query to calculate the total number of successful and failure mission outcomes

```
%sql select "Mission_Outcome", count(*) from SPACEXTABLE group by "Mission_Outcome";
```

- There were 100 successful launches in total, and 1 failure.
- This query selects the number of instances of each mission outcome from the table, grouped by the mission outcome.

- Query result:

Mission_Outcome	count(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- I used the following query to identify names of boosters that have carried the maximum payload:

```
%sql select distinct("Booster_Version") from SPACEXTABLE where
"PAYLOAD_MASS__KG_" = (select MAX("PAYLOAD_MASS__KG_") from
SPACEXTABLE);
```

- This query selects the unique booster names where the payload mass is equal to the maximum, which was identified using a subquery to the table.

- Query result:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- I used the following query to identify the month names, failure landing_outcomes in drone ship, booster versions, launch_site for the months in year 2015:

```
%sql select Booster_Version, Launch_Site, Landing_Outcome,
substr(Date, 0, 5) as Year, substr(Date, 6,2) as Month from
SPACEXTABLE where Year = '2015';
```

- This query selects the relevant columns using substrings to extract the year and month from the date column filtered to only return results from 2015.

- Query result:

Booster_Version	Launch_Site	Landing_Outcome	Year	Month
F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)	2015	01
F9 v1.1 B1013	CCAFS LC-40	Controlled (ocean)	2015	02
F9 v1.1 B1014	CCAFS LC-40	No attempt	2015	03
F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)	2015	04
F9 v1.1 B1016	CCAFS LC-40	No attempt	2015	04
F9 v1.1 B1018	CCAFS LC-40	Precluded (drone ship)	2015	06
F9 FT B1019	CCAFS LC-40	Success (ground pad)	2015	12

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- I used the following query to identify landing outcomes in the date range 2010-06-04 to 2017-03-20:

```
%sql select Landing_Outcome, count(Landing_Outcome) as Number from
SPACEXTABLE where Date between '2010-06-04' and '2017-03-20' group
by Landing_Outcome order by Number desc;
```

- This query selects the landing outcomes and their counts (grouped by outcome) from the table between the dates requires and presented in descending order.

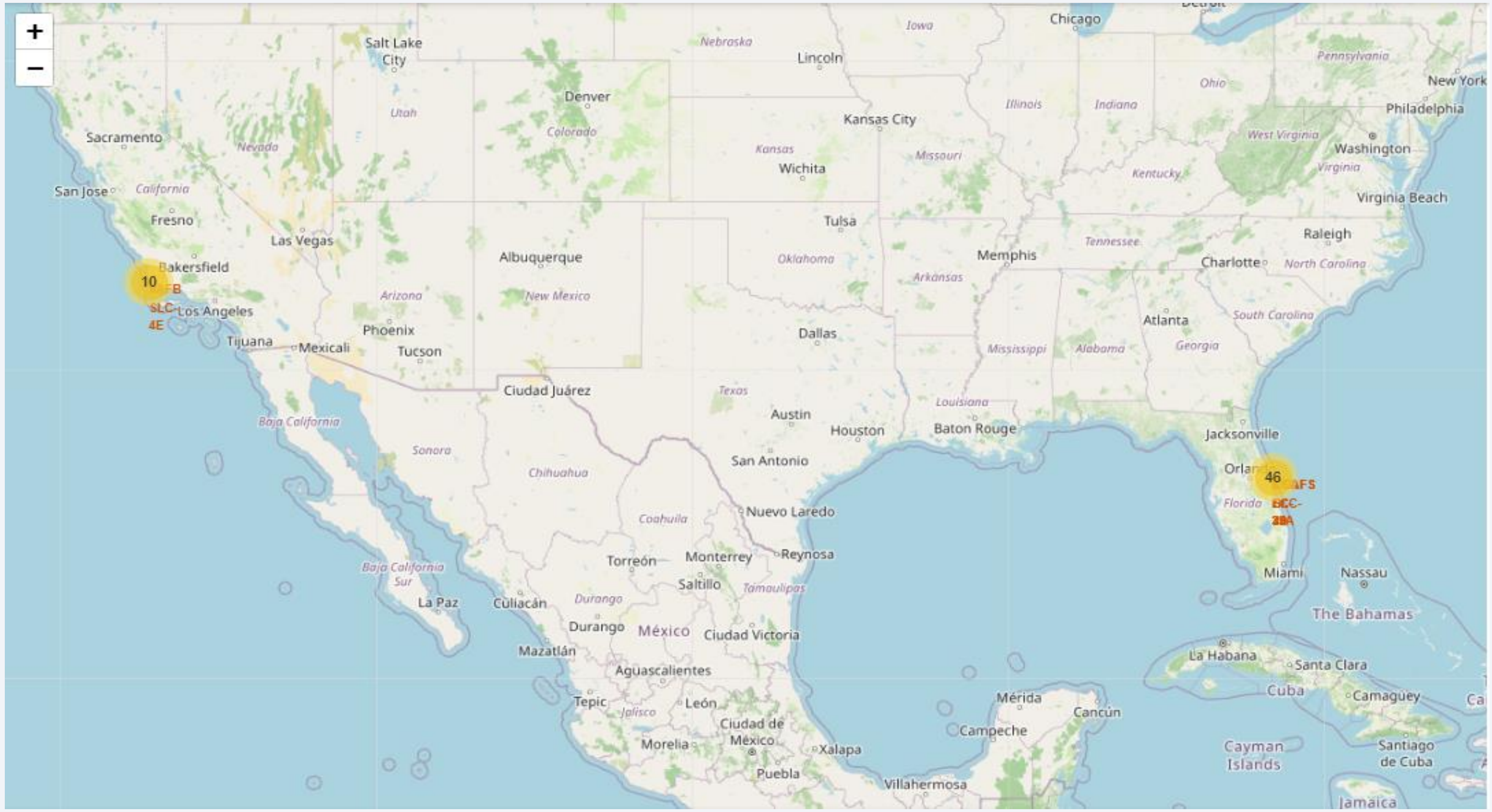
- Query result:

Landing_Outcome	Number
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

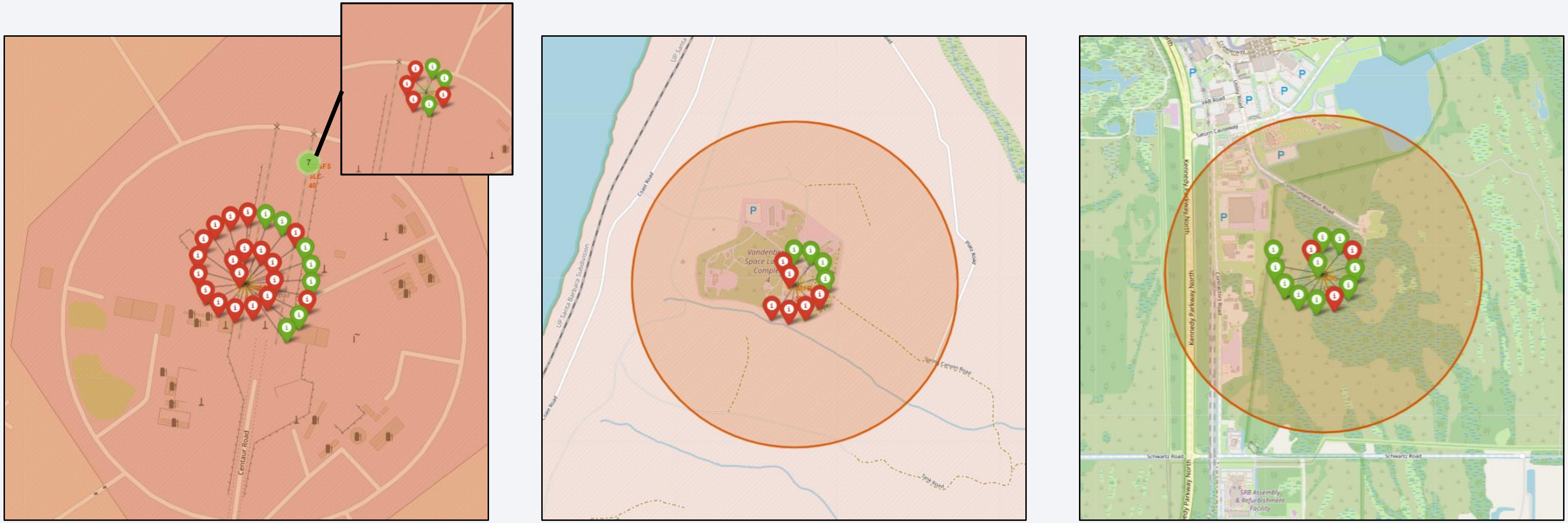
Launch Sites Proximities Analysis

Map of Launch Site Locations



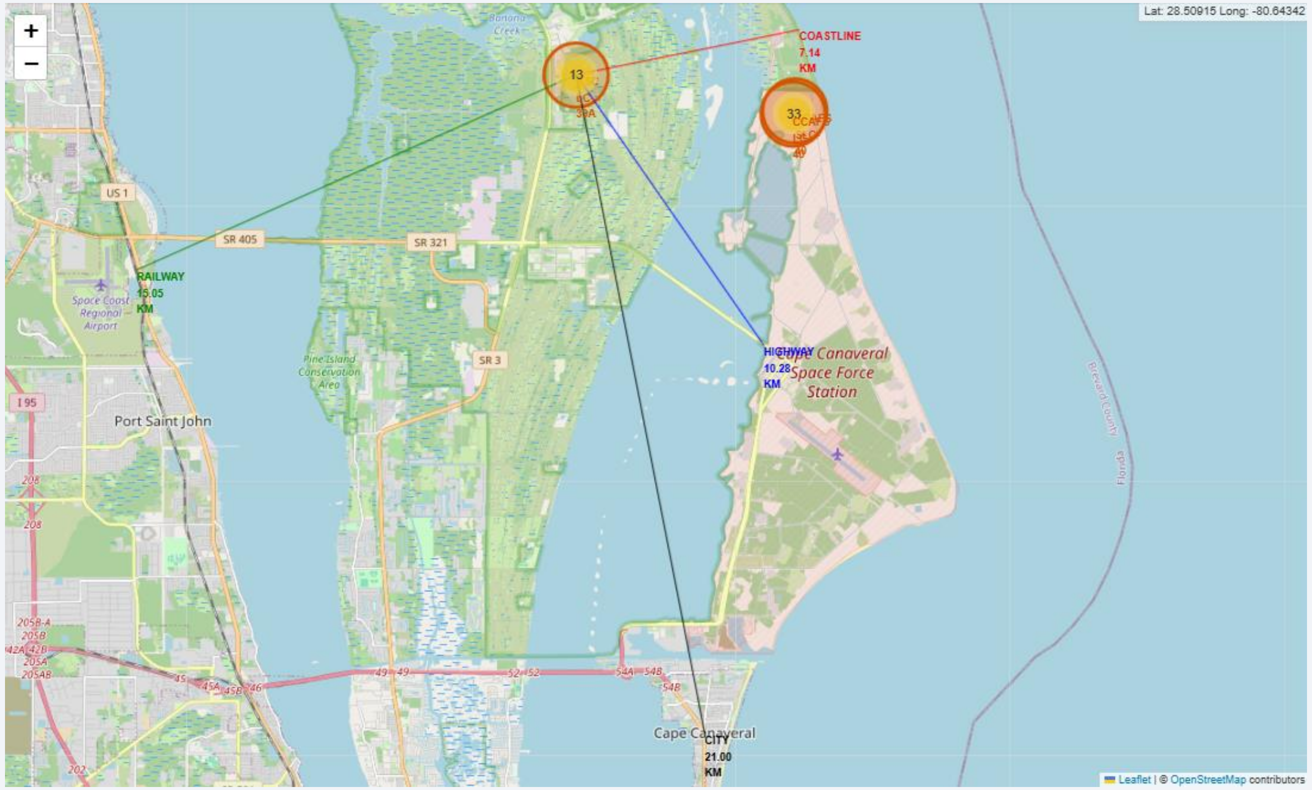
- Launch sites (red dots with names as labels) and the launch clusters (yellow circles with number of launches as the number label). The three sites are located close to the equator within the continental United States and on its coastline.

Map of Launch Outcomes across all Sites



- Map of launch outcomes from CCAFS SLC 40, VAFB SLC 4E, and KSC L39A (left to right), denoted by the red shaded circular region. Green markers denote successful launches and red markers denote failed launches.

Map of Landmarks Nearby Kennedy Spaceflight Centre

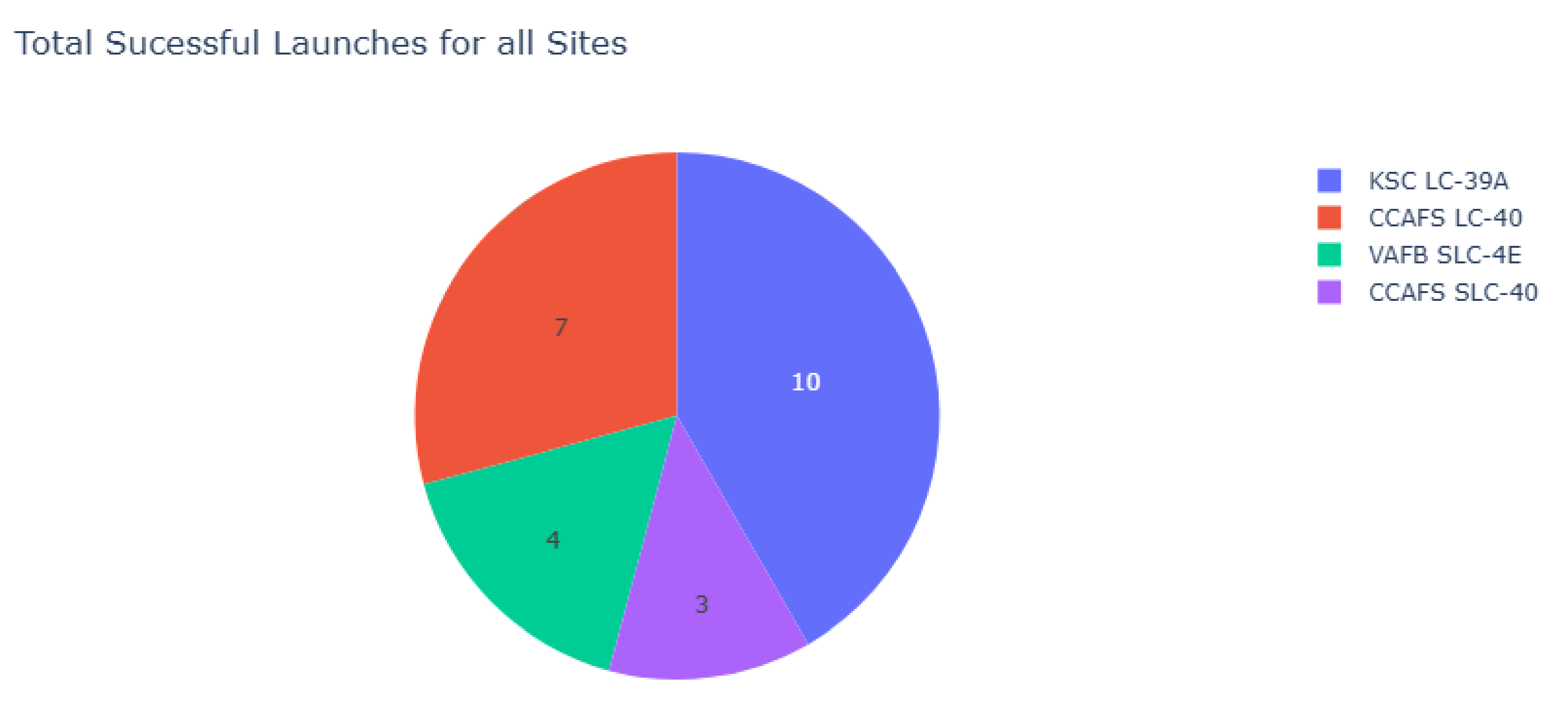


- Markers showing launch sites (red circles) and their respective clusters (yellow circles) and the environs of KSC L39A including Cape Canaveral city (black, 21.00km south), A railway line (green, 15.05km west), the coastline at False Cape (red, 7.14 km east), and a highway (blue, 10.28 km south-east).

Section 4

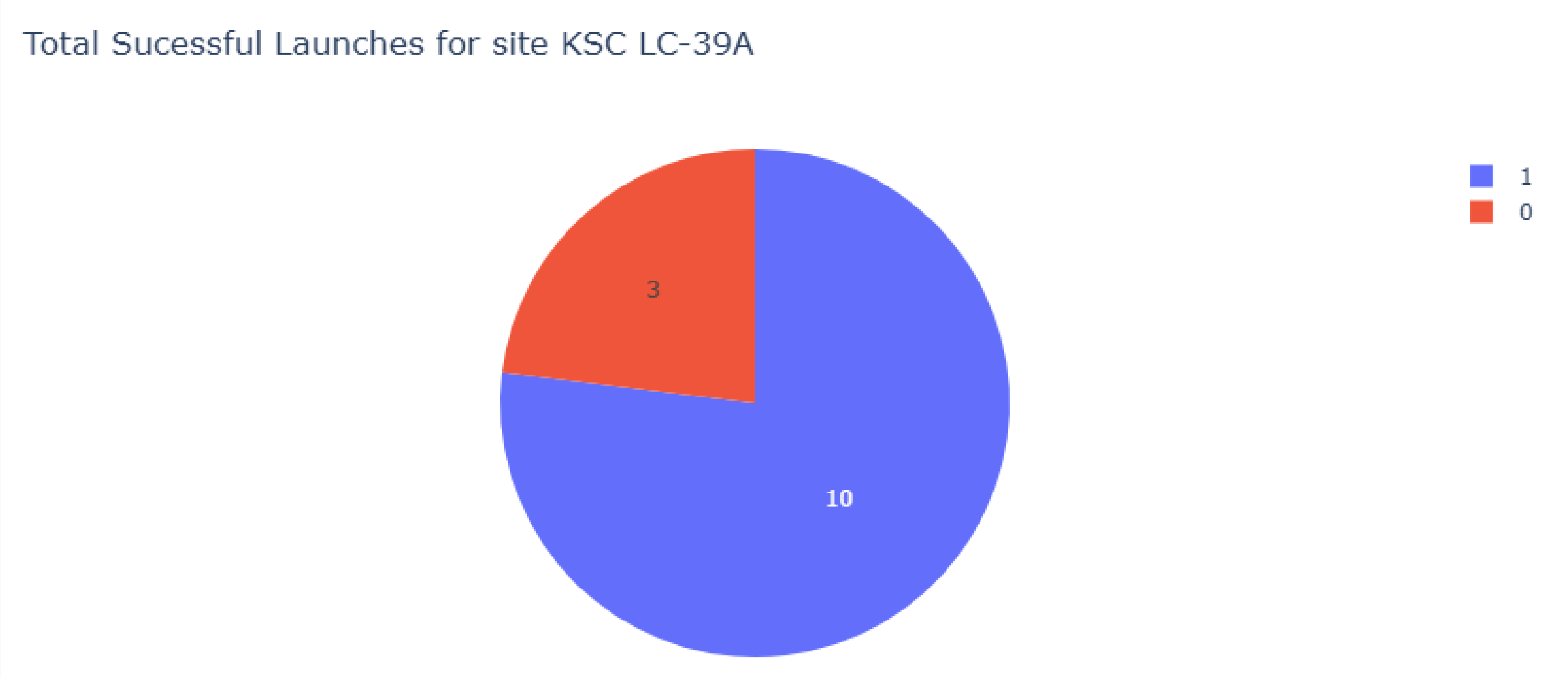
Build a Dashboard with Plotly Dash

Launch Success for all Sites



- The plot shows a pie chart illustrating the number of successful launches for each site, individually noted by coloured wedges with the number of successful launches as a numeric value within the respective wedge. A legend in the top right corner shows which colour denotes which launch site.

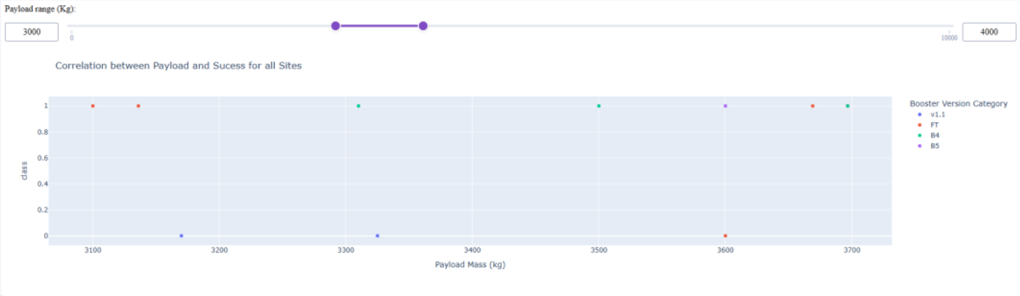
Launch Success for Site KSC LC-39A

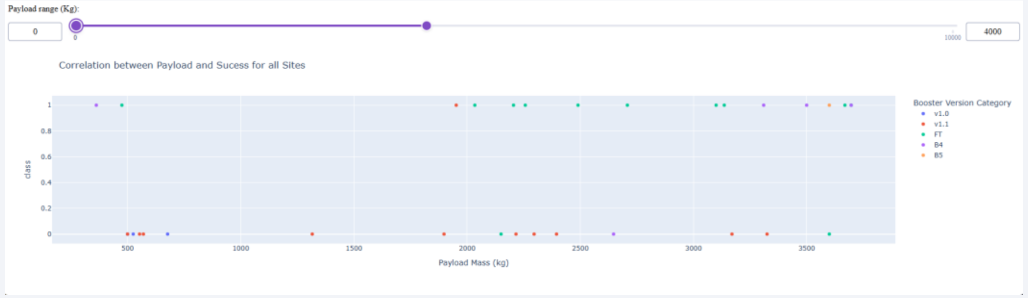


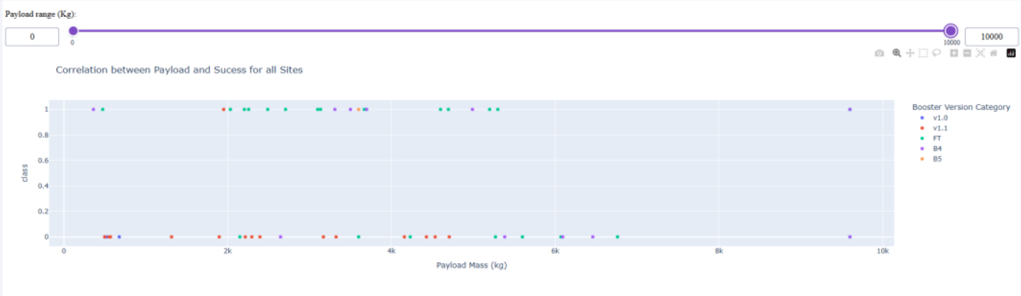
- The plot shows a pie chart illustrating the number of successful launches (in purple, class '1') and failed launches (in red, class '0') for the KSC LC-39A launch site. A legend denotes the colour key for the two classes.

Payload versus Launch Outcomes for all Sites

- The top plot shows the outcomes for payloads in the range 3000 to 4000 kg, which shows the greatest success, with 7 out of 10 successful launches.
- The middle plot shows the outcomes for payloads in range 0 to 4000 kg. Most of the failures in this range are early block v1.1 boosters (10/15 failures).
- The bottom plot shows the outcomes for the full range of payload masses 0 to 10000 kg. The rate of failure increases as payload increases above 5500 kg.







Section 6

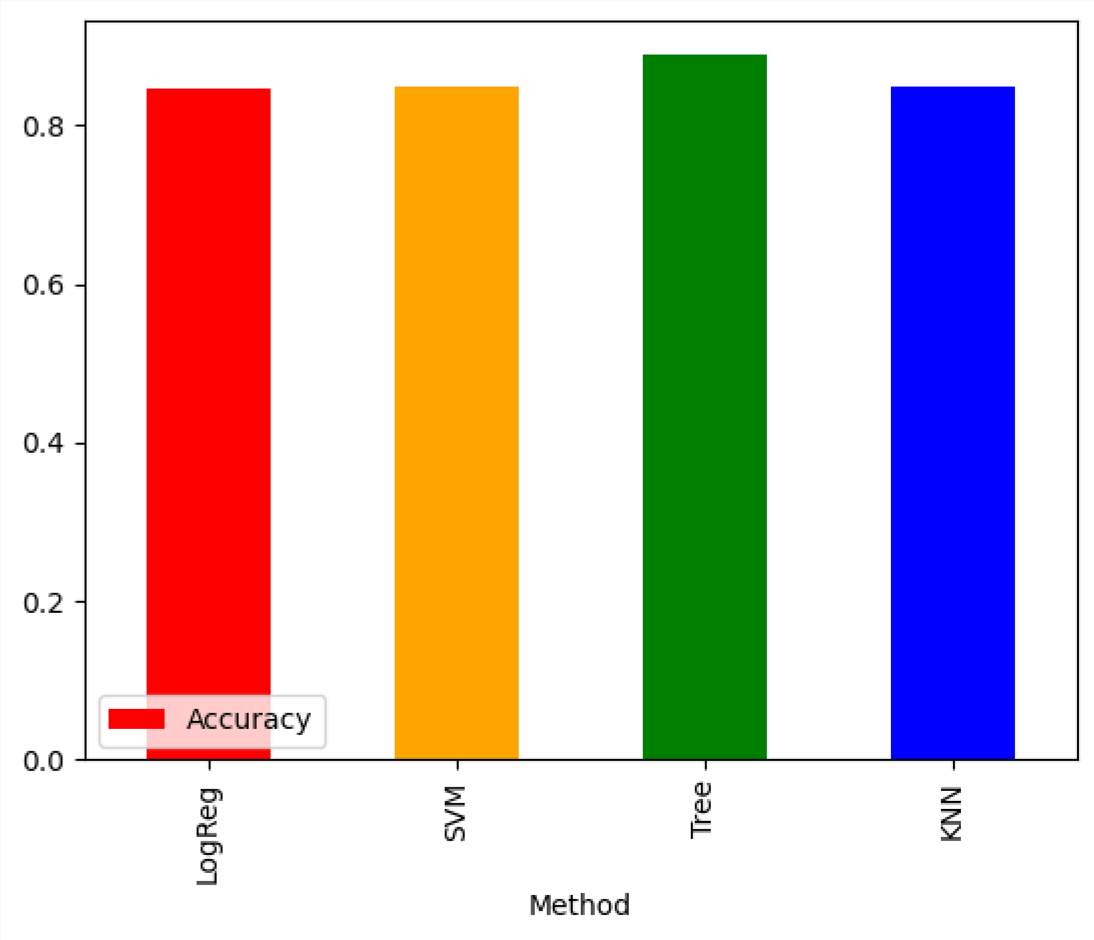
Predictive Analysis (Classification)

Classification Accuracy

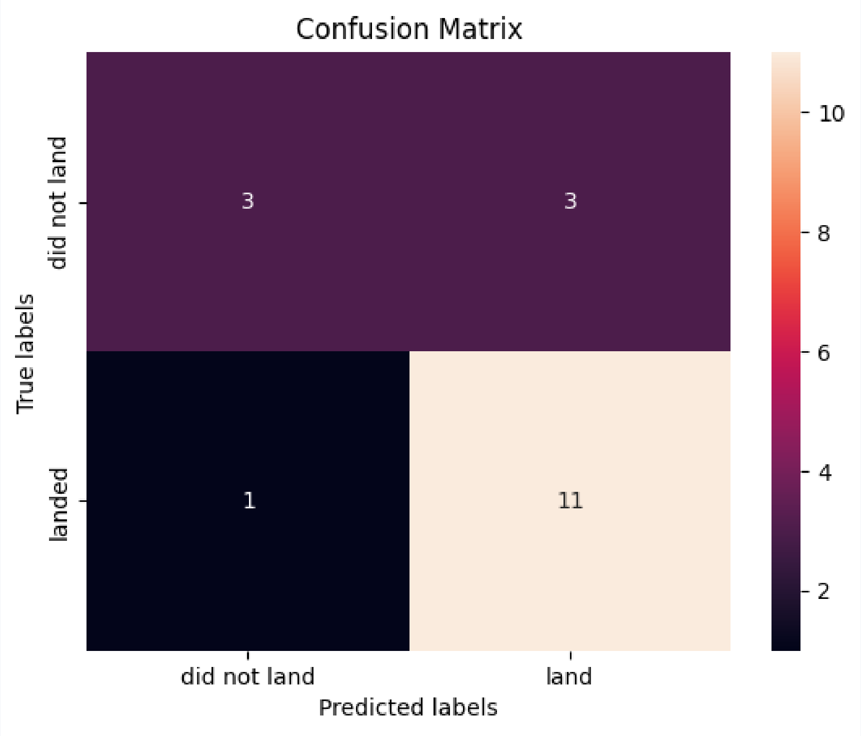
- The Decision Tree Classifier method has the highest accuracy at 89%.

- Results:

	Method	Accuracy
0	LogReg	0.846429
1	SVM	0.848214
2	Tree	0.887500
3	KNN	0.848214



Confusion Matrix



- The above confusion matrix for the Decision Tree method shows strong diagonality such that the predicted labels (x-axis) align well with the true labels (y-axis). There is one false negative (bottom left) and three false positives (top right).

Conclusions

- Rocket launch sites are located in coastal areas closest to the equator within the continental United States. These sites are served by substantial local infrastructure, e.g. railways, highways, but are distant from local population centres. The KSC L39A launch site has the greatest overall success rate of 77%.
- The success rate of rocket launches increases over time pointing to increasing reliability. Launch success for SpaceX in 2020 is around 80%, having grown from 0% in 2013.
- The payload mass of rocket launches increases over time, and the height of orbit increases over time, both pointing to increasing technical capability.
- The most reliable predictor of a successful booster landing is the Decision Tree Classifier model, with an accuracy of 88%.

Appendix

- The reduced data, analysis, and results of this project can be found in the collected files, Python scripts, and Jupyter notebooks held at: https://github.com/jontymarshall/IBMDataScienceCapstoneProject/

Thank you!